

Release Note for HiSIM_HV 2.5.1

Hiroshima University

Revision: HiSIM_HV 2.5.1

Date: 2023.02.09

Deliverable List of HiSIM_HV 2.5.1

Deliverable	File Name or Directory Name	Date
Manual	HiSIM_HV 2.5.1_Users_Manual.pdf	2023.02.09
Release Note	HiSIM_HV 2.5.1_Release_Note.pdf	2023.02.09
Default Model Parameters	HiSIM_HV 2.5.1_Default_Model.txt	2023.02.09
Default Instance Parameters	HiSIM_HV 2.5.1_Default_Instance.txt	2023.02.09
Default Model Parameters (CODEP=3)	HiSIM_HV 2.5.1_Default_Model_CODEP3.txt	2023.02.09
Verilog-A Code	HiSIM_HV 2.5.1_VA-Code/	2023.02.09
Educational Community License v2.0	LICENSE.txt	2023.02.09
Notice of Support by Si2 and Release Date	NOTICE.txt	2023.02.09

Update information of HiSIM_HV 2.5.1 from HiSIM_HV 2.5.0

- Added a flag on body-bias clamping for the intrinsic MOSFET model.
Added flag: **COVBSCL = 1** (Default). **COVBSCL = 0** inactivates the bias clamping.
- Added a flag on body-bias clamping for the overlap capacitance model.
Added flag: **COQOVCL = 1** (Default). **COQOVCL = 0** inactivates the bias clamping.
- User-defined maximum and boundary values in common with the two kinds of bias clamping options.
Added parameters: **VBFWDMX = 0.8** (Default) and **VBFWDBND = 0.4** (Default).
- Code improvement on the calculation of the overlap charge to avoid possible numerical difficulties around the flatband condition for the gate overlap region.
- Removal of the special bypass procedure near around the gate threshold voltage for the continuity of Ids.
- Bug-fix for glitches in Ids under very low Vds condition (Pds_ini is updated).
- Local protections of Pds for the CLM model (Lred) and the high-field mobility model (Mu).
- Bug-fix to avoid FPE in the power function of negative base value (Eyd for the thermal noise model).
- Incorporated the Vds mode into the flicker noise for the correct Pnoise simulation.
- Changed the discipline file name from “discipline.h” to “disciplines.vams”.
- Code cleaning to pass VAMPyRE v1.8.7 (ERROR/WARNING/NOTICE/STYLE/hidden states) for hisimhv.va, hisimhv_n4.va and hisimhv_n5.va. There are three ERRORS left due to the second derivatives in the induced gate noise model. Five NOTICES find five macros never used, but those are to be used for the aging model when _AGING_ is defined. (If “ifdef” is introduced here, one of the major simulators fails in Verilog-A compilation even with its latest version.)
- Bug-fix of Vds-mode handling in the simplified overlap charge model (Qgos & Qgod).
- Incorporated the FD condition into the analytical solution (initial guess) of the surface potential for the overlap charge model.
- Numerical protections for the unclamped bias (-Vxbgmt) case in the overlap charge model.

- 15) Fully iterative Ps0LD is introduced as a new option of the model flag **COQOVSM** to solve the discontinuity in overlap capacitance at its flat-band. **COQOVSM** = 4 activates this model option.
- 16) Bug-fix of Qbu overwritten in the deep PT model: Qbu is localized not to affect AC characteristics.
- 17) Numerical protection in the deep PT model: Argument of sqrt gets negative because of (Vbipn - Vbscl).
- 18) Extension of the deep punchthrough (PT) current model: Added a model option with a constant potential value of **PS0PT**, especially for the negative drain voltage condition.
Added parameter: **PS0PT** = 0 (Default). **PS0PT** > 0 activates this model option.
- 19) Revised the equation of the potential barrier height in the deep PT model. This update eliminates the Vds dependence of the deep PT current under the positive high drain voltage condition.